Lehong Wang

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Education

Carnegie Mellon University (CMU) — Pittsburgh, PA, USA
Master of Science in Mechanical Engineering, Research track — *Aug. 2024 – May. 2026*

Worcester Polytechnic Institute (WPI) — Worcester, MA, USA
Bachelor of Science in Robotics Engineering and Computer Science (double major) — *Aug. 2020 – May. 2024*

Related Courses: Deep Learning, Machine Learning, Software Engineering, Data Structures and Algorithms, Discrete Differential Geometry, Web Development, Robotics, Control, Motion Planning, Embedded Computing

Technical Skills

Skill Sets: Software Development, Deep Learning, Machine Learning, Computer Vision, Robotics, Research, 3D Reconstruction, Agentic AI, Computational Geometry, CAD Design, Robot Arm Programming, Motion Planning, Control, Additive Manufacturing, Finite Element Analysis, Mechatronic System Development, Cloud Computing, 3D Art

Programming Languages: Python, MATLAB, C++, C, JavaScript, Java

Engineering Tools: Pytorch, OpenCV, Docker, Colab, AWS, GCP, CUDA, LangChain, Ollama, WebGL, Flask, ROS, Rviz, CGAL, Gazebo, Solidworks, Arduino, Blender, Unreal Engine, COMSOL, R studio, KiCAD, Fusion360, CCSstudio

Languages: English (fluent), Chinese/Mandarin (native)

Internship Experience

Computer Vision Internship @ Shelfmark (https://shelfmark.com) — May. 2025 – Aug. 2025
Graduate Research Intern — *Pittsburgh, PA, USA*
- Designed and implemented software framework for 3D geometry reconstruction and industrial inspection.
- Investigated sota learning-based algorithms to efficiently extract accurate 3D point cloud information from images.
- Built algorithm, workflow and hardware to reconstruct high-res pointcloud using photometric and stereo methods.
- Led inspection projects with industry clients, deploying defect detection systems in factory production facilities.

Internship @ SeeM (https://seemuseums.art) — Nov. 2024 – Apr. 2025
Research Intern — *Pittsburgh, PA, USA*
- Researched and developed software to build 3D reconstruction of museum artifacts from mobile phone videos.
- Experimented with latest novel view synthesis algorithms and deep learning models to obtain desirable results.
- Assisting in the deployment and optimization of the 3D reconstruction pipeline as cloud-based services on AWS.
- Contributed to deployment of AI agents that provide interactive, fact-verified descriptions of museum artifacts.

Projects

Conformal Additive Manufacturing — May. 2023 – Present
Research team leader (10-member research project) @ Biorobotics Lab — *Pittsburgh, PA, USA*
- Developed software framework for transforming 2D PCB design onto arbitrary high curvature 3D surface.
- Implemented mesh generation, parametrization, and point cloud reconstruction algorithms for motion generation.
- Designed and Programed UR robot arm and motorized platform for conformal FDM on high-curvature surfaces.
- Led the mechatronic and software development of high-precision motion system for multi-axis fabrication.
- Assist with digital twin data collection and Manufacturing Futures Institute (MFI) Digital Backbone integration.

Siemens Electroic Assembly Process Tracking Project — May. 2025 – Present
Research Assistant (6-member research project) @ Biorobotics Lab — *Pittsburgh, PA, USA*
- Developed integrated system for Siemens' battery assembly division to monitor and verify human assembly.
- Designed and implemented safety-aware robot arm and camera system for collaborative human–robot assembly.
- Built human action recognition software that interprete and provid corrective feedback on assembly procedures.

Anchor Alignment Adaptive Weighting for Robust Domain Generalization — Dec. 2024 – Apr. 2025
Research Project — *Worcester, MA, USA*
- Developed novel algorithm leveraging NLP anchors to enhance domain generalization and noise robustness.
- Designed a weighted loss function that dynamically adjusts sample contributions based on NLP-derived anchors.
- This work is under review at IEEE Transactions on Knowledge and Data Engineering (TKDE).

Automated Defect Detection in PCB Manufacturing — Aug. 2024 – Dec. 2024
Team Member (4-member teamwork project) — *Pittsburgh, PA, USA*
- Implemented and improved a SSD-based model for defect detection and classification for PCB images.
- Implemented classical architectures like Transformer, BiLSTM, and Resnet for face and voice recognition.
- Built a custom torch-like library that implemented Transformer, LSTM and CNN from scratch with numpy.

Quadruped with Robot Arm for Loco-Manipulation Tasks — Aug. 2023 – May. 2024
Team member (6-member teamwork capstone research project for undergraduate degree) — *Worcester, MA, USA*
- Collaborated in a team of 6 members to develop software and hardware add-ons for the Unitree Go1 robot.
- Researched and modified quadruped motion planning and perceptive locomotion frameworks, such as OCS2.
- Implemented modified framework with perception on the Go1 robot and demonstrated improved locomotion.

Fluidic Circuit Auto-generation Software — May. 2022 – May. 2024
Research Assistant @ Robotic Materials Group — *Worcester, MA, USA*
- Developed a design software customized for the automation of fabrication-ready fluidic circuitry design.
- Incorporated software into lab component of graduate course and consistently enhanced it based on user feedback.
- **Best Paper Award** at 2024 RoboSoft conference; additional paper accepted at Advanced Robotics Research.

Vision-based Close-loop Printing for Desktop FDM Printers — May. 2023 – Nov. 2023
Team Leader (4-member teamwork project) @ Robotic Materials Group — *Worcester, MA, USA*
- Led the team to develop software that analyzes printing gcode, detects defects, and generates fixing commands.
- Implemented the approach on a desktop FDM printer and successfully improved the quality of the print.
- This work was published at 2024 IEEE Conference on Soft Robotics (RoboSoft) as **Oral Presentation**.